

Theory of Random Forests: A Review

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Abstract

Random forests (RF) have a long history, originally defined by Leo Breiman in 2001, but with antecedents in bagging methods introduced in 1996. They have become one of the most widely adopted machine learning tools thanks to their computational efficiency, relative insensitivity to tuning parameters, in-built cross validation, and interpretation tools. Despite their popularity, mathematical theory about the fundamental properties of random forests has been slow to emerge. Nonetheless, the past decade has seen significant advances in our understanding of all these processes. In this review paper, we describe several variations of RF and how rates of consistency of these variants highlight the impact of different RF mechanisms on their performance. Another line of research focuses on establishing central limit theorems and confidence intervals for random forests. We also depict recent analyses in variable importance computed with RF.

Keywords: Random forests, excess risk, central limit theorem, variable importance.

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1 INTRODUCTION

Decision trees are among the most comprehensible machine learning methods: a recursive partition of the input space is constructed from the data, and the prediction at a new point results only from an average (or majority vote) of the observations falling within the same leaf as that point. The instability of these methods in the face of dataset perturbation and their limited predictive capacity have led to the consideration of tree-based ensemble methods such as random forests or gradient boosting decision trees. Despite the advent of deep learning, such methods are part of the state of the art for dealing with tabular datasets. What’s more, these ensemble methods benefit from the versatility of trees: they can handle a variety of problems (classification, regression, quantile estimation, survival...), missing data of different types and are computationally attractive. For a long time, the statistical properties of random forests remained a mystery. Before delving into the various directions of random forest research, we start by describing the original algorithm.

1.1 Notations and algorithm

Throughout the paper, we assume to be given a dataset $\mathcal{D}_n = \{(X_i, Y_i), i = 1, \dots, n\}$, composed of i.i.d. pairs (X_i, Y_i) distributed as the generic pair (X, Y) , where $X \in \mathbb{R}^d$ and $Y \in \mathbb{R}$. This general setting includes continuous and discrete inputs X and covers classification and regression problems.

At its most general, a random forest consists of an average of M prediction decision trees, which we write as

$$\hat{f}(x) = \frac{1}{M} \sum_{m=1}^M T(x; \omega_m, \mathcal{D}_n), \quad (1)$$

in which T is the prediction at $x \in \mathbb{R}^d$ of a decision tree trained on the data $\mathcal{D}_n$ along with independent and identically distributed randomization parameters ω_m . Because the $T(x; \omega_m, \mathcal{D}_n)$ for $m = 1, \dots, M$ are i.i.d. conditional on the data, it is common to consider the limit $M \rightarrow \infty$ yielding the *infinite forest* (see [Scornet et al., 2015](#))

$$\hat{f}_\infty(x) = \mathbb{E}_\omega T(x; \omega, \mathcal{D}_n), \quad (2)$$

where the expectation is taken w.r.t. ω only. Indeed, M can be made large enough to remove variation associated with ω at the price of additional computing.

A random forest is thus defined by two choices: the decision tree algorithm and how the randomization parameters ω_m impact the tree structure/prediction. The original random forest algorithm was designed by Breiman (2001), who proposed aggregating fully-developed CART (specific decision trees, see Breiman et al., 1984) randomized via data resampling and with, at each node, feature subsampling. Since then, the class of random forest-like algorithms has significantly expanded, with variations proposed to improve performance, to expand the applicability of random forests to new tasks or data types, and to produce algorithms that are theoretically tractable or have desirable distributional properties. Here we outline the broad structure of a forest and note the choices that were made in Breiman’s original version; many of the results that we detail below are given for specific variations. Note that almost all random forest variants are based on sequentially-chosen diadic splits, that is, the feature space $\mathbb{R}^d$ is recursively divided into cells, until some stopping criterion is satisfied.

- *Data resampling.* The original random forest algorithm obtained bootstrap samples from $\mathcal{D}_n$ to create an input $\mathcal{D}_n^*$ to each tree which we characterize using the vector N_{im} indicating the number of times observation i is repeated in the sample given to the m th tree. However, most theoretical advances have been made by selecting a random subset of size k either with (Mentch and Hooker (2016), Wager and Athey (2018)) or without (Scornet (2016a), Zhou et al. (2021)) replacement. It is also possible to rely solely on randomization within the tree-building process, as is implicitly done in defining the equivalent kernels described in Scornet (2016b).
- *Split randomization - Set of potential splits.* Most random forest algorithms consider partitioning the current cell based on the value of one of the dimensions of X . Early tree building algorithms such as CART Breiman et al. (1984) considered the midpoint between each successive data value, along each dimension, as potential splits. Random forests restricted these to a randomly chosen subset of `mtry` dimensions. More restrictive versions are limited to the medians of the observed data values or the midpoints of the current node range; others restrict the imbalance in the proportion of data on each side of the split. Other variants consider splits based on linear combinations of features.
- *Tree building process - Best split selection.* How to choose between potential splits selected above? Here random forests follows CART in scoring each split by the decrease in squared prediction error when the prediction on each side of the split is held constant (see, e.g., Geurts et al., 2006); many extensions consider criteria based on particular models. *Honest* forests (Wager and Athey, 2018) choose among splits based on 1/2 of the data available to the current tree, reserving the other half to select values in the leaves. Other random forest variants choose randomly among candidate splits.
- *Tree building process - Stopping rule.* The original random forests ceased splitting regions with fewer than k observations (with $k = 1$ or 5 by default). Other models can specify tree depths, trees pruned in the manner of CART, or stochastically chosen stopping rules.
- *Tree prediction.* The original random forests return a constant value which minimizes the loss (squared error, or weighted misclassification cost) in each leaf. *Honest* forests use a held-out set of data to decide on this value to keep the leaf value independent

from the structure of the tree. Other algorithms estimate parametric models in each leaf (see, for example, Chaudhuri et al., 1994).

- *Prediction combination.* The original random forest algorithm was designed as a prediction tool and produced either an average, if predicting a real-valued quantity, or a majority vote if a class label was needed. Averaging remains the most common way to combine the predictions from trees, but weighted averages, or using leaf indicators within a larger regression model (Agarwal et al., 2023) are also possible.

Figure 1 presents a schematic of random forests algorithms. Note that other descriptions are possible: for instance, the selection of data can be described as a process internal to the tree. However, the collection of algorithmic choices presented above encompasses most proposed practical random forest variants, and their mathematical models.

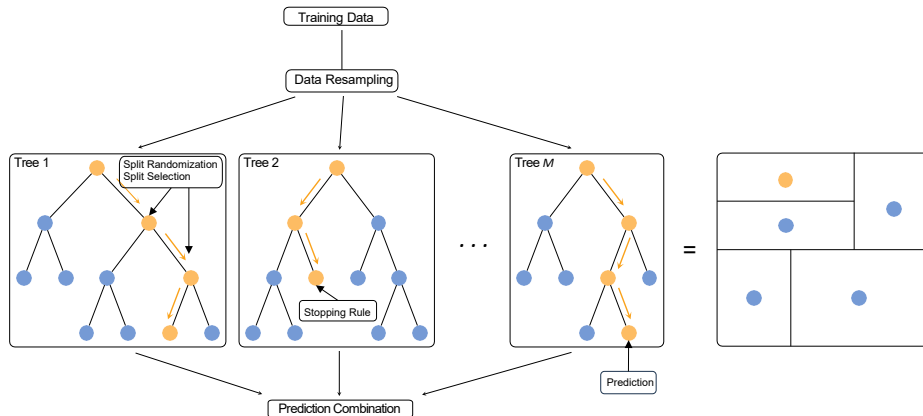


Figure 1: Illustration of a random forest on the left and a tree partition on the right.

In this review, we focus on four different aspects of random forest theory. In Section 2, we detail different rates of consistency for several random forest models, with the goal of elucidating the role of each RF hyperparameter on RF predictive performance. In Section 3, we review variants of RF that have been theoretically studied. In Section 4, we detail central limit theorems for random forests and the resulting uncertainties quantifications. In Section 5, we summarize the analyses of variable importances in random forests, in order to provide guidance on how to use them. Finally a discussion of further research avenues is presented in Section 6.

Several earlier papers already review some aspects of random forests theory. Notably, Biau and Scornet (2016) provide a detailed review of the theoretical properties of random forests at that time. From a methodological perspective, Loh (2014) provide a very detailed review of various decision tree algorithms. A description of different applications that can be solved using random forest (or its variants) are provided in Criminisi et al. (2012). Boulesteix et al. (2012) describe the existing implementations of random forests, with an emphasis on advances in bioinformatics. The interested reader is also referred to Probst et al. (2019) for insightful discussions on the impact of parameter tuning in random forests. Thus, the present review focuses on the most recent theoretical advances in this field.

2 BIAS, VARIANCE AND CONSISTENCY

The analysis of random forests is notoriously difficult due to a combination of several ingredients – recursive (tree) structure with data-dependent splits, split/data randomization, large tree depths, aggregation – each element of which produces theoretical challenges. As first attempts to dissect their internal behavior, several research works designed new stylized versions of random forests, whose splitting procedure does not depend on the data. Such forests, called *purely random forests* (see [Arlot and Genuer, 2014](#)), or non-adaptive random forests, allow us to comprehend the contribution of some key ingredients (split randomization, tree depth) on the predictive performance of the forest. The strength of the results comes at the price of simplification of the original algorithm. Another route consists in deriving results for Breiman’s random forests: results focus on simpler properties, while requiring much more involved technical tools. We explore these two avenues in the following sections. We will work with the general regression model

$$Y = f^*(X) + \varepsilon, \quad (3)$$

where $X \sim \mathcal{U}([0, 1]^d)$, $f^*(X) = \mathbb{E}[Y|X = x]$ is the regression function, ε a noise satisfying $\mathbb{E}[\varepsilon|X] = 0$ with variance $\mathbb{V}[\varepsilon|X] \leq \sigma^2$ a.s. In this section, all results are stated for infinite random forests (see [\(2\)](#)). They can be extended to finite random forests if the number of trees M is chosen large enough.

2.1 Purely random forests

The first purely random forest is the centered random forest in which each cell of each tree is built in the following way: first, a direction is selected uniformly at random among $\{1, \dots, d\}$, and then the split is made at the center of the cell, along the chosen direction. The process is repeated until a prespecified depth is reached, which results in balanced trees. Centered random forests were first introduced by [Breiman \(2004\)](#) and later analyzed by [Biau \(2012\)](#) who derived an upper bound on their rate of consistency. According to Theorem 5 in [\(Biau, 2012\)](#), assuming that f^* is Lipschitz, the centred forest $\hat{f}_{k,n}^{\text{cen}}$ of depth k satisfies

$$\mathbb{E} \left[(\hat{f}_{k,n}^{\text{cen}}(X) - f^*(X))^2 \right] \lesssim n^{-1/((4/3)d \log 2 + 1)}, \quad (4)$$

with $k \sim C_d \log_2(n)$ and $C_d = (1 + 3/(4d \log 2))^{-1}$. This rate was further improved by [Klusowski \(2021\)](#) (Theorem 2), which proves that, under similar assumptions,

$$\mathbb{E} \left[(\hat{f}_{k,n}^{\text{cen}}(X) - f^*(X))^2 \right] \lesssim n^{-\frac{1}{d \log 2 + 1} (1 + \Delta_d)}, \quad (5)$$

where $\lim_{d \rightarrow \infty} \Delta_d = 0$. [Klusowski \(2021\)](#) showed that this rate was not improvable for general Lipschitz functions. Recall that the minimax rate over the class of Lipschitz functions is given by $n^{-2/(d+2)}$ ([Stone, 1982](#)). Many estimators reach this optimal rate, among which kernel methods and simple cubic partitions (regular partitions, a very-specific tree form). Thus, it may seem surprising that randomizing tree partitions leads to a suboptimal estimator, as highlighted by [5](#). In fact, the slow rate of convergence of centered forest results from their (relative) bad approximation capacities ([Klusowski \(2021\)](#)): the bias term is of order $2^{-k/(d \log 2)}$, whereas it is of order $2^{-2k/d} < 2^{-k/(d \log 2)}$ for a single cubic partition. Therefore, the randomness of the tree partition (and in particular the fact that all cells do

not have the same shape) harms the approximation error of the forest compared to a single cubic partition.

While thorough analyses established the exact properties of centered forests, studying such models does not allow for a precise understanding of Breiman’s forests. The most important limitation of centered forest is the data-independent splitting process, which makes the algorithm unable to select only relevant variables, thus not adapting to sparsity. Besides, the analysis does not show any benefits from using centered forests instead of centered trees, whereas in practice, Breiman’s forests outperform individual CART trees. Since lower and upper bounds match (see [Klusowski, 2021](#)), this is the result of an overly simplistic model of centered forests rather than from an improvable theoretical analysis. Finally, the rate of consistency is optimal when leaves contain numerous observations, whereas original random forests build fully-developed trees (whose leaves contain few observations)¹.

2.2 Uniform forests

Uniform random forests, introduced by [Biau et al. \(2008\)](#) work as centered forests, except that split position is drawn uniformly at random instead of being made at the middle of the cell. The resulting tree is still balanced of depth k . [Biau et al. \(2008\)](#) prove the consistency of uniform forests when the depth k tends to infinity, such that $n/2^k$ also tends to infinity, thus ensuring a growing number of observations per cell on average. [Arlot and Genuer \(2014\)](#) (Equation 53) show that the risk of such forests satisfies

$$\mathbb{E} \left[(\hat{f}_{k,n}^{\text{uni}}(X) - f^*(X))^2 \right] \lesssim n^{-2\alpha/(2\alpha+1)}, \quad (6)$$

with $\alpha = -\log_2(1 - 1/(2d))$. It can be shown that the risk of a single uniform tree is larger than the risk of the forest. However, uniform forest do not reach the minimax rate of convergence for C^2 functions, as $2\alpha/(2\alpha + 1) < 4/(4 + d)$. [Arlot and Genuer \(2014\)](#) claim that this slow rate of convergence results from the fact that all cells are split equally, even if some are much smaller than others (see also the comment above on centered forests and [Klusowski, 2021](#)).

Contrary to previous analyses, [Arlot and Genuer \(2014\)](#) show that the number of trees must be larger than $M = (1 - 1/2d)^{-k}$ to ensure that the predictive performances of the finite forest (with M trees) is close to that of the infinite forest.

2.3 Mondrian Forests

Splitting schemes that take into account the geometry of the current partition can be considered instead of centered or uniform forests. Such partitions are still independent of the dataset but contain cells designed to be of similar shapes. An example of such a construction is the Mondrian forest ([Lakshminarayanan et al., 2014](#)), which originates from Mondrian processes ([Roy et al., 2008](#)). In Mondrian forests, the probability of splitting a cell is proportional to its perimeter, then a direction is chosen at random, proportionally to the side length and the position is drawn uniformly, along the chosen direction. A lifetime parameter controls the depth of the tree. Since leaves of Mondrian trees have shapes close to hypercubes, such forests exhibit an approximation error of the same order as cubic partitions for

¹However, note that according to the optimal expression of k_n in [4](#), in a high-dimensional setting, centered trees are almost fully-grown.

Lipschitz function, and are thus minimax over this class (see [Mourtada et al., 2020](#)). Moreover, contrary to a single Mondrian tree, or a cubic partition, Mondrian Forests are also minimax over the class of Holder functions with exponent $\beta \in (1, 2)$ (see [Mourtada et al., 2020](#)). Hence, there is an improvement of Mondrian forest compared to a single Mondrian tree in terms of rate of consistency. Such phenomenon results solely from the randomization of partitions.

2.4 Median Forests

In Breiman’s forests, two types of randomization occur: a randomization of partitions due to split randomization and subsampling, the latter inducing changes in the optimal split positions, and a randomization of labels due to subsampling only; see [Arlot and Genuer \(2016\)](#) for a thorough discussion. We have so far considered randomization of partitions due to split randomization only, as purely random forests do not use the dataset to split. In order to analyze all randomizations at work in random forest, we now turn to the median forests. Median trees (see Section 20.3 in [Devroye et al., 2013](#)) are built by uniformly choosing a split direction and then cut at the empirical median of the inputs along this direction. The resulting cells contain the same number of points up to ± 1 observation. As this construction depends on the position of the dataset inputs, we can build fully-developed trees whose leaves contain exactly one observation, bringing us closer to the original random forest algorithm. As for any fully-developed non-adaptive tree, a single median tree is inconsistent (see Problem 20.2 in [Devroye et al., 2013](#)). However, median forests, composed of fully-developed median trees, are consistent if either subsampling ([Scornet, 2016a](#), [Duroux and Scornet, 2018](#)) or split randomization ([Arnould et al., 2023](#)) is properly incorporated into the model.

[Duroux and Scornet \(2018\)](#) show that tuning the subsample size a_n or the tree depth k_n in median forests results in the same upper bound on the rate of consistency (improved later by [Klusowski, 2021](#)). This suggests that these two hyperparameters play the same role in median forests and thus, only one of them needs to be optimized (see also [Tang et al., 2018](#), for the importance of subsampling). More importantly, their analysis can be carried out in the exact same way if split randomization is removed with all trees choosing the same split directions, thus stressing the impact of subsampling on forest (rate of) consistency. On the other hand, disabling subsampling, and studying only split randomization for fully-developed median trees, [Arnould et al. \(2023\)](#) prove that the risk of such median forests satisfies

$$\mathbb{E} \left[(\hat{f}_n^{\text{med}}(X) - f^*(X))^2 \right] \lesssim (\log_2 n)^{-(d-1)/2}. \quad (7)$$

These results highlight the benefits of split randomization on the consistency of the forest. However, the rate of consistency in (7) is very slow. It turns out that such a phenomenon is expected for fully-developed non-adaptive random forests. Indeed, [Lin and Jeon \(2006\)](#) proved that the risk of a non-adaptive forest trained on the full dataset, is larger than

$$\frac{(\log n)^{-(d-1)}}{n_{\text{obs/leaf}}}, \quad (8)$$

where $n_{\text{obs/leaf}}$ is the number of samples inside each leaf of each tree. Therefore, non-adaptive fully-developed forests that do not incorporate subsampling can’t reach rates of consistency faster than $(\log n)^{-(d-1)}$. Such results seem to point toward the use of subsampling instead of split randomization in non-adaptive forests.

2.5 Slight variants of Breiman’s forest

Wager and Walther (2015) define (α, k) -valid tree partitions that are (i) constructed on the full original sample without bootstrapping or subsampling, (ii) whose splits leave at least a fraction α of parent node observations in each child node, and (iii) such that each final leaf contains at least k observations. Standard implementations of random forests allow removing bootstrap step (i) and a minimum number of observations in each leaf (iii). Condition (ii) is not satisfied in general, as it is known that splits performed along noise features are close to the edges of the cell (End-Cut Preference, see Ishwaran, 2015, Breiman et al., 1984). Enforcing (ii) thus requires to modify the splitting criterion. However, the overall algorithm remains quite close to the algorithm used in practice.

Wager and Walther (2015) establish a control on any random forest using (α, k) -valid tree partitions. More precisely, assuming that $\liminf d/n > 0$ and under regularity constraints on the regression model,

$$\mathbb{P} \left[\sup_{x \in [0,1]^d, \Lambda_1, \dots, \Lambda_M} |\hat{f}_{\Lambda_1, \dots, \Lambda_M}(x) - f_{\Lambda_1, \dots, \Lambda_M}^*(x)| \lesssim \sqrt{\frac{\log(n) \log(d)}{\log((1-\alpha)^{-1})}} \frac{1}{\sqrt{k}} \right] \rightarrow 1, \quad (9)$$

where $\hat{f}_{\Lambda_1, \dots, \Lambda_M}(x)$ is the prediction of the forest composed of the tree partition $\Lambda_1, \dots, \Lambda_M$ and $f_{\Lambda_1, \dots, \Lambda_M}^*(x)$ is the same forest where empirical means in each tree partition are replaced by conditional expectation of the output variable in the corresponding cell. Based on the above concentration result, Wager and Walther (2015) propose the Guess-and-Check forest and prove its pointwise consistency.

Wager and Athey (2018) introduce the honesty property: a tree is *honest* if each observation cannot be used simultaneously for the tree construction and for estimating the mean response in terminal nodes. A simple manner for a tree to be honest is to use a subsample for the tree construction and the rest of the observations for mean leaf estimation. Under regularity conditions, Wager and Athey (2018) prove the pointwise consistency of honest trees with (α, k) -valid partitions in the context of causal inference.

2.6 Breiman’s forests

What may be the most difficult ingredient of Breiman’s forests is the data-dependent splitting criterion: both inputs and outputs are used to find the optimal splits via exhaustive search. The process does not involve any gradient-based strategies, and thus theoretical analyses cannot leverage the vast literature on gradient descent. Such adaptive partitions are intrinsically difficult to analyze, as they do not correspond to a local averaging estimate that can be addressed using Stone’s theorem since weights depend on the labels, see Stone (1977).

Scornet et al. (2015) is the first work to study the original Breiman’s algorithm. Considering an additive model of the form

$$Y = \sum_{j=1}^d f_j(X^{(j)}) + \varepsilon, \quad (10)$$

the original random forest was proved to be consistent, assuming that the number of leaves t_n tends to infinity while satisfying $t_n = o(a_n/(\log a_n)^9)$, where a_n is the subsample size. Due to the previous condition, the consistency result of Scornet et al. (2015) is not valid

for fully-developed trees ($t_n = a_n$). They also establish the ability of random forests to asymptotically identify the relevant direction for splitting in a sparse additive framework.

Using other proof techniques, [Klusowski and Tian \(2022\)](#) extend these results to additive (logistic and ordinary) regression, establishing oracle inequalities on the excess risk of generic decision trees. They also prove that the rate of consistency for Breiman’s random forests is upper bounded by $(\log n)^{-1}$. [Klusowski and Tian \(2022\)](#) highlights that split randomization cannot damage the predictive performances of RF by more than a factor 3 (with default RF parameters). [Klusowski and Tian \(2022\)](#) also prove the consistency of RF in sparse high-dimensional settings, where the dimension d is allowed to grow sub-exponentially with n and where additive components can be discontinuous (Corollary 7.2). Assuming a minimum impurity decrease in all tree nodes, [Chi et al. \(2022\)](#) derive upper bounds on the approximation error of random forests in high-dimensional settings, highlighting the influence of the number of direction eligible for splitting. In the case of some specific high-dimensional r -sparse regression models with binary input features, [Syrkanis and Zampetakis \(2020\)](#) establish rate of consistency of MSE of Breiman’s forests of order $2^r \log(d) \log(n)/n$, which increases logarithmically with the ambient dimension d .

While decision trees are able to detect signal and are thus tailored for sparse settings, they fail to adapt to the simplicity of additive models. Using rate-distortion theory, [Tan et al. \(2022\)](#) establish a lower bound on the MSE of general honest decision trees, implying that they fail to recover the efficient rate of $n^{-2/3}$ of Generalized Additive Models ([Stone, 1985](#)). We note that while these results bring us closer to Breiman’s original forests, they still rely on modifications such as subsampling and controlling tree depth. The impact of these choices relative to the original algorithm remains unclear.

2.7 Interpolation, self-regularization and double descent

Recent works ([Belkin et al., 2019](#)) show that overparameterized interpolating estimators do not automatically overfit the data. This seems to contradict the common belief, based on the bias-variance tradeoffs, stating that model complexity must be reasonable, in order to prevent variance from exploding. Such a phenomenon, called *double descent* was established for overparameterized neural networks and for random forests in [Wyner et al. \(2017\)](#), [Belkin et al. \(2019\)](#). Double descent phenomenon is observed for random forest when the complexity of the model is controlled via the number of leaves per tree *and* the number of trees (see Figure 5 in [Belkin et al., 2019](#)). In the binary classification setting, even in presence of bootstrap, random forest interpolate exactly the training set. [Curth et al. \(2024a\)](#) argue that the double descent phenomenon only occurs when the complexity is measured via two distinct quantities – the number of leaves and trees – merged into one single axis. Unfolding these two quantities leads to classical U-curve (see Figure 1 and 2 in [Curth et al., 2024a](#)).

[Wyner et al. \(2017\)](#) advocates that RF good predictive performances come from their *spiked-smooth* nature (see Figure 1 in [Wyner et al., 2017](#)), which results directly from their interpolating properties. However, [Mentch and Zhou \(2020\)](#) claim that interpolation in itself is not sufficient to explain the good performances of RF, as bagging (without split randomization) has lower predictive performances while still interpolating in binary classification setting. According to [Mentch and Zhou \(2020\)](#), RF good performances result from the feature randomization (via `mtry`), which acts as implicit regularization: low values of `mtry` leads to better predictive performances in low SNR (Signal-To-Noise) settings. In the same setting, [Mentch and Zhou \(2022\)](#) show that adding input noise variables can improve RF predictive performance. In fact, tuning the tree depth instead of considering

fully-grown trees in RF may act as a beneficial regularization in low Signal-to-Noise Ratio settings (Zhou and Mentch, 2023).

Curth et al. (2024b) propose to reconcile these two views. The “spiked-smooth” behavior mentioned by Wyner et al. (2017) is quantified for RF by Curth et al. (2024b) via an *effective parameters* measure: the *effective parameters* value computed on test points is lower than that computed on training points, which provides empirical evidence of a spiked RF behavior near training points and a smooth behavior far from training points. The effective parameters value computed on test points also decreases when the number of trees, the feature randomization (small values of `mtry`), or the tree depth increase (see Figure 4 in Curth et al., 2024b). Thus, the good performances of RF seem to result of their “spiked-smooth” behavior, something that is exaggerated with interpolating trees or low values of `mtry`.

3 VARIATIONS

Beyond algorithmic modifications for purposes of theoretical tractability, or elucidating the role of computational mechanisms, many variations on random forests have been proposed on the grounds of improving performance, with the purpose of providing new functionality, or repurposing some aspects of random forests. We provide a brief catalog of some of these variants here, particularly focusing on recent methods – see Loh (2014) for earlier variants.

3.1 Performance Improvements

RF improvements come either from theoretical or heuristic justification. Some of these modify splitting criteria used for constituent trees. Tomita et al. (2020) splits along randomly selected sparse linear combinations of features, while Hornung and Boulesteix (2022) uses two-variable combinations to split out axis-oriented subsets of the current node. These splits are a strict superset of axis-oriented splits and necessarily increase the expressive capacity of trees, but the additional complexity increases the challenges of a theoretical motivation. Cattaneo et al. (2024) analyses a variant of these; with additional constraints on both the regression function, the tree partition and the number of observations per node, oblique RF can reach a rate of consistency of $n^{-2/(2+q)}$, where q acts as an effective dimension. Alternative splitting criteria fit models within nodes; Raymaekers et al. (2024) develops computationally fast methods to include a further linear term at each split, choosing between enforcing continuity across the split or allowing a break.

Other variants focus on specific tasks or model structures; Basu et al. (2018) explicitly uses the structure of random forests to emphasize high-dimensional interactions, iteratively reweighting data, and re-fitting forests, based on the data’s participation in interactions defined through strings of important splits within a tree. Fink et al. (2010) proposes a method aimed at ecological survey data in which each tree is obtained from data that is localized to one of many overlapping spatiotemporal regions. This induces interactions between time and location and any landscape or ecological features used in the data, in particular preventing the forest from using relationships in one ecological context in contexts that could be quite different. Coleman et al. (2019) employ a reweighting strategy when covariate shift is expected at the time of deployment.

Further extensions use random forests within larger models. Tan et al. (2025) extends Frieman and Popescu (2008) in performing regression on indicators for the leaves of each

tree with linear functions. Ghosal and Hooker (2020) “boosts” random forests; fitting a second random forest to the out-of-bag residuals of the first. Bayesian methods for trees and ensembles of trees (e.g. Chipman et al., 2012, Jeong and Rockova, 2020) do not explicitly use random forest structures, but averages of posterior samples create forest-like averages; see Ročková and Van der Pas (2020) for posterior consistency of these ensembles.

3.2 Functionality

The original Random Forests was developed for regression and classification tasks, but this has been significantly expanded. A first strategy for these is to adapt splitting criteria to reflect the data model. Here, Saha et al. (2021), Rabinowicz and Rosset (2022) assume spatially correlated predictions and split based on log likelihood. Ishwaran et al. (2008) develops random forests for survival analysis; accounting for censoring, with consistency studied in Cui et al. (2022). Meinshausen and Ridgeway (2006) developed a random forest quantile regression. There has also been considerable interest in causal estimation; Wager and Athey (2018) adapted the tree building process to estimate conditional average treatment effect, extended to a survival setting in Cui et al. (2023). Athey et al. (2019) presents a general model for splits maximizing a likelihood with approximate optimization for models in each daughter node to speed computation.

3.3 Secondary Uses

The structure of trees in random forests has also been employed to provide an adaptive kernel or distance function to be used in further models. In the context of purely random trees, Scornet (2016b) developed an explicit connection between random forests and kernel methods. This distance is induced by the frequency over trees with which an observation falls within the same leaf as the point of interest; these are often referred to as “potential nearest neighbors” (Lin and Jeon, 2006). When the tree structure is obtained from data, its adaptivity results in a kernel that, heuristically, locally mimics the contours of the target of interest, although we are aware of few results studying this. This adaptivity has been used to produce prototypes for classification (Tan et al., 2020), as the kernel in a local linear regression (Friedberg et al., 2020) who demonstrates an improvement in the resulting convergence rate, and to provide prediction intervals based on out-of-bag residuals (Lu and Hardin, 2021) or conduct more formal quantile regression Li et al. (2024).

4 CENTRAL LIMIT THEOREMS AND VARIANCE ESTIMATION

While much of the analysis of convergence rates rests on the analysis of tree building algorithms, providing uncertainty quantification in terms of confidence intervals or variance estimates for $\hat{f}_n(x)$ has generally focused on the ensembling structure of the random forest. Here it will be helpful to explicitly re-write the subsampling step into the random forest (1):

$$\hat{f}_n(x) = \frac{1}{M} \sum_{m=1}^M T(x; \omega_m, D_n^{I_m}),$$

in which I_m is an a -vector of the indices $1, \dots, n$ and $D_n^{I_m}$ selects the corresponding set of observations $(X_{i_1}, Y_{i_1}), \dots, (X_{i_a}, Y_{i_a})$ Brieman’s original forest employed a bootstrap sample,

generating $a = n$ observations sampled with replacement, but employing subsampling with or without replacement has proved valuable for variance estimation as well as consistency.

Early approaches to estimating $\text{var}(\hat{f}_n(x))$ centered around variations of the bootstrap or jackknife for ensemble methods. [Sexton and Laake \(2009\)](#) employed a “bag of little bootstraps” approach: generating B ensembles $\hat{f}_n^1(x), \dots, \hat{f}_n^B(x)$, each of M/B trees resulting in a final prediction $\frac{1}{B} \sum_{b=1}^B \hat{f}_n^b(x)$ along with uncertainty obtained via the standard deviation among the $\hat{f}_n^b(x)$ along with a correction for the bias of the standard deviation; the consistency of this estimate was later demonstrated by [Athey et al. \(2019\)](#).

In a similar manner, methods based around the infinitesimal jackknife were developed by examining the infinite forest [\(2\)](#). The infinitesimal jackknife operates by attaching equal weights to each observation and examining the *directional derivative* U_i – the gradient of the estimate with respect to the weight given on observation i . For ensemble methods, [Efron \(2014\)](#) attached weights in the form of selection probabilities when generating I but not for how observations are used subsequently, this also reflects the implementation of weights within the `randomForest` package in R ([Liaw and Wiener, 2002](#)). The application of this to random forests is given in [Wager and Walthers \(2015\)](#), where the resulting direction derivatives are given by

$$U_i = \text{cov}_I(T(x; D_n^I, \omega), N(I)_i),$$

where $N(I)_i$ indicates the number of times observation i is repeated in sample I . For a finite forest, the theoretical covariance above is simply replaced with the empirical covariance over the subsamples I_m . We then obtain an estimate

$$\widehat{\text{var}}(\hat{f}_n(x)) = \sum_{i=1}^n U_i^2$$

although this can be substantially biased unless M is much larger than usually employed in practice.

Beyond variance estimates, inferential tools like confidence intervals also require distributional results. For random forests, these have been obtained by observing a connection between an infinite forest and the classical construction of U-statistics ([Hoeffding, 1948](#)) when the sets I_m are obtained from subsamples taken without replacement. [Mentch and Hooker \(2016\)](#), [Wager and Athey \(2018\)](#), [Peng et al. \(2022\)](#) developed versions of a central limit theorem extending basic U-statistics constructions. To formally state these results, we examine a forest of M trees obtained using subsamples of size a_n and define the Hájek projection

$$g_c((x_1, y_1), \dots, (x_c, y_c)) = \mathbb{E}[T(x; D_n^I, \omega) | (X_1, Y_1), \dots, (X_c, Y_c)]$$

to be the expected value of T conditioned on the first c data points with g_{a_n} understood to also include ω , and define the corresponding variance components

$$\zeta_{c, a_n} = \text{var}[g_c((X_1, Y_1), \dots, (X_c, Y_c))].$$

Then, so long as $\lim_{n \rightarrow \infty} (a_n/n)(\zeta_{a_n, a_n}/a_n \zeta_{1, a_n}) = 0$ and the response Y is appropriately regular, we have

$$\frac{\hat{f}_n(x) - \mathbb{E}[\hat{f}_n(x)]}{\sqrt{a_n^2 \zeta_{1, a_n}/n + \zeta_{a_n, a_n}/M}} \xrightarrow{d} N(0, 1).$$

The limit condition above reflects a heuristic that $\zeta_{a_n, a_n} \propto a_n \zeta_{1, a_n}$ reducing the condition to $a_n = o(n)$. Note here that the centering in this theorem is given by $\mathbb{E}[\hat{f}_n(x)]$ rather than a target of the form $\mathbb{E}[Y|X = x]$. This is because the construction of the result ignores the tree-building process and cannot guarantee that the bias in $\hat{f}_n(x)$ is ignorable. In that sense, this result cannot be used to guarantee that confidence intervals derived from it are consistent. One can argue that the stability of the forest is of interest independently of formal coverage guarantees. Alternatively, [Wager and Athey \(2018\)](#) do demonstrate a CLT centered on $\mathbb{E}[Y|X = x]$ by a careful control of both the tree-building process and choosing a subsampling rate $a_n = n^{-\beta}$ for a range of β that depend on constraints on the asymmetry of allowable splits. This results in a general $a_n/n = n^{-1+\beta}$ convergence rate. [Athey et al. \(2019\)](#) demonstrate the asymptotic equivalence of generalized random forests to a U -statistics with equivalent statistical properties; similar analyses are carried out for random survival forests in [Cui et al. \(2023\)](#).

[Peng et al. \(2022\)](#) also provided a Berry-Esseen theorem for complete U -statistics, that is assuming that M is large enough to be ignorable, resulting in:

$$\sup_{z \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\hat{f}_n(x) - \mathbb{E}[\hat{f}_n(x)]}{\sqrt{a_n^2 \zeta_{1, a_n}/n}} < z \right) - \Phi(z) \right| < \frac{6.1 \mathbb{E}[|g_1|^3]}{\sqrt{n} \zeta_{1, a_n}^{3/2}} + (1 + \sqrt{2}) \left(\frac{a_n}{n} \left(\frac{\zeta_{a_n, a_n}}{a_n \zeta_{1, a_n}} - 1 \right) \right)^{1/2}$$

Here for fixed subsample sizes a_n , the bound has a classical $n^{-1/2}$ rate, but specific rates in the infinite-order case depend on the scaling of the variance components $(a_n \zeta_{1, a_n}, \zeta_{a_n, a_n})$; as for the CLT above assuming these remain bounded, $a_n = o(n)$ suffices for the bound to vanish asymptotically.

Within the context of estimating variances, both the infinitesimal jackknife and bootstrap-of-little-bags can be shown to be consistent for $\zeta_{1, n}$. However, both employ the variance over a Monte Carlo approximation to quantities defined for an infinite forest – either a forest built on a bootstrap sample, or $\text{cov}(T, N(I))$; this was also the case for the two-level estimate of [Mentch and Hooker \(2016\)](#). As such, the estimated variance is inflated by the Monte Carlo variance of the estimate, and this bias can be substantial unless much larger forests are built than are needed for good prediction. While intuitive estimates of the Monte Carlo variance are available, they are themselves inflated by the without-replacement sampling scheme which induces negative correlation between trees, conditional on the data. [Zhou et al. \(2021\)](#) instead examined a forest built using samples taken *with* replacement in which subsamples can be viewed as simulations from the empirical distribution and provided a general estimation scheme, defining

$$\begin{aligned} \mu_i &= \sum_{m=1}^M N_{im} T(x; D^{I_m}, \omega_m) / \sum N_{im}, \quad i = 1, \dots, n \\ SS_\tau &= \sum_{i=1}^n \sum_{m=1}^M N_{im} (\mu_i - \bar{\mu})^2, \quad SS_\epsilon = \sum_{i=1}^n \sum_{m=1}^M N_{im} (T_m - \mu_i)^2 \\ \hat{\zeta}_{1, a_n} &= \frac{SS_\tau - (n-1)SS_\epsilon / (Ma_n - n)}{Ma_n - \sum_i (\sum_m N_{im})^2 / (Ma_n)} \end{aligned}$$

Here the μ_i are the means of trees for which observation i was included in the subsample, weighted by its inclusion frequency and the resulting $\hat{\zeta}_{1, a_n}$ is unbiased for data simulated from the empirical distribution. The use of subsamples with replacement moves forests from U -statistics to being V -statistics. The classical equivalence between these (e.g. [Van der](#)

Vaart, 2000) only holds when $a_n = o(n^{1/4})$ (Shieh, 1994); Zhou et al. (2021) re-expresses a general V statistic as a U statistic with a modified kernel with the equivalent central limit theorem, and the variance estimates above remain consistent and unbiased for data generated from the empirical distribution.

The distributional results defined here only characterize the variability of $\hat{f}_n(x)$; they do not provide prediction intervals. These can be obtained by quantile regression (Meinshausen and Ridgeway, 2006), or Lu and Hardin (2021) developed an out-of-bag form of conformal inference (Angelopoulos et al., 2023). Bias corrections have also been studied in Hooker and Mentch (2018), Lu and Hardin (2021), Ghosal and Hooker (2020) these all add a correction $\hat{g}_n(x)$ to $\hat{f}_n(x)$ which improved predictive performance on a range of regression problems, often very significantly. Ghosal et al. (2025) demonstrate the extension of infinitesimal jackknife (IJ) methods to estimate the covariance between models to which IJ is applicable and hence to iterative bias corrections of this form. Finally, note that these distributional results rest entirely on the subsampling process of random forests rather than the specifics of tree construction. They thus apply more generally to any bagged estimator. They also apply to any quantity calculated from individual trees and then averaged over the forest; that includes all of the variable importance measures studied below.

5 VARIABLE IMPORTANCE

The original implementation of random forests come with two importance measures, widely used in practice: the Mean Decrease in Impurity (MDI) and the Mean Decrease in Accuracy (MDA). We review below their theoretical properties and improvements that have been proposed recently. Although relatively rarely employed, central limit theorems for these measures follow immediately by repurposing the distributional theory for random forests and observing that each is a quantity defined from individual trees that is then averaged over the forest², thus allowing uncertainty quantification.

5.1 Mean Decrease in Impurity (MDI)

The MDI (Breiman, 2002) of any variable $X^{(j)}$ is defined for each tree of the forest as the weighted sum of the impurity reduction in each node that splits along the j th variable, the weights being simply the fraction of data points falling into each node. More formally, the MDI of the j th variable, computed via a tree T is defined as

$$\widehat{\text{MDI}}_T(X^{(j)}) = \sum_{\substack{A \in T \\ j_{n,A}=j}} p_{n,A} L_{n,A}(j_{n,A}, z_{n,A}),$$

where $L_{n,A}(j_{n,A}, z_{n,A})$ is the impurity reduction in the cell A and $p_{n,A}$ the fraction of data points falling into A . The MDI of a forest is then simply the average of all MDI computed via the trees that compose it.

Louppe et al. (2013) established one of the first expression of the population version of the MDI, for fully-grown trees built on categorical covariates. In this context, a tree splits on all possible covariates and deriving the MDI expression reduces to enumerating all

²MDA can be defined with respect to a generic model and when implemented generically does not take this form, however Breiman (2001) defined variable importance as an average over trees, taken using out-of-bag samples.

eligible splits. In this scenario, [Louppe et al. \(2013\)](#) proved that the sum of the MDI on all input variables is equal to the mutual information of Y and $X^{(1)}, \dots, X^{(d)}$, highlighting that MDI is a natural importance measure that is spread through all input variables. [Louppe et al. \(2013\)](#) also establish that the importance of irrelevant features (independent of Y conditional on any subset of covariates) is zero. While interesting, the setting presented in [Louppe et al. \(2013\)](#) does not permit to study the impact on MDI performances of split positions (all variables are categorical and split so that each node corresponds to a modality) nor tree depth (trees are fully grow).

The CART splitting procedure is more likely to select variables with a large number of split points (as continuous features or categorical variables with a lot of values). As a consequence, the MDI is positively biased towards such variables (see, e.g., [Nicodemus, 2011](#), [Boulesteix et al., 2012](#)). Such a phenomenon was first noticed by [Breiman et al. \(1984\)](#) (see also [Strobl et al., 2007](#)). Similarly, MDI of variables containing many missing values may be overinflated (see, e.g., [Kim and Loh, 2001](#)).

In order to reduce these intrinsic biases, several works proposed to replace the usual splitting procedure in CART by statistical tests, such as F-test on ordered variables (FACT, [Loh and Vanichsetakul, 1988](#)), chi-squared tests on categorical ones (see algorithms QUEST and CRUISE in [Loh and Shih, 1997](#), [Kim and Loh, 2001](#)), permutation tests, or tests on the maximal value of Gini criterion ([Strobl et al., 2007](#)). To the best of our knowledge, there exist no detailed theoretical analyses of these procedures.

Even in the absence of dependencies between inputs and outputs, variables with many categories are more likely to be split ([Zhou and Hooker, 2021](#)). This phenomenon is related to multiple testing problems, since the quality of many split points are evaluated to choose the best one. Because a unique dataset is used to build the tree and evaluate the MDI, the resulting MDI may be positively biased for such high cardinality variables. In order to remove this bias, [Zhou and Hooker \(2021\)](#) propose using two different datasets to respectively build the tree and compute the MDI. [Li et al. \(2019\)](#) propose another way to use an extra-sample based on a rewriting of the MDI of any variable $X^{(j)}$ as

$$\frac{1}{n} \sum_{i=1}^n f_{T,j}(X_i) Y_i, \quad (11)$$

where

$$f_{T,j}(x) = \sum_{A \in T, j_A=j} (\bar{Y}_{A_L} \mathbb{1}_{x \in A_L} + \bar{Y}_{A_R} \mathbb{1}_{x \in A_R} - \bar{Y}_A \mathbb{1}_{x \in A}). \quad (12)$$

Equation (11) shows that MDI can be seen as a covariance between the output and a function $f_{T,j}$ related to the tree partition. [Li et al. \(2019\)](#) propose to use a part of the data to build the partition (and thus $f_{T,j}$) and the other to estimate the MDI via Equation (11), employing similar motivation to that of honesty in [Wager and Athey \(2018\)](#).

Deep trees tend to produce biased MDI. Indeed, [Scornet \(2023\)](#) proves that the sum of all MDI produced by a fully-grown tree tends to the mutual information of Y and $X^{(1)}, \dots, X^{(d)}$ plus the variance of the model noise. Any input variable can have their MDI modified by an additive factor of the noise variance. Such phenomenon highlights the asymptotic empirical bias of fully grown trees. A finite sample analysis of the same phenomenon was proposed by [Li et al. \(2019\)](#). Consider a model with d input variables, whose noise variables – features

assumed independent of all other quantities – are indexed by $k \in \mathcal{S} \subset \{1, \dots, d\}$. For any tree T , let

$$G_0(T) = \sum_{j \in \mathcal{S}} \widehat{\text{MDI}}_T(X^{(j)}) \quad (13)$$

be the sum of importances of noise variables. Then, letting $\mathcal{T}_n(m_n, k_n)$ be the set of decision trees whose leaves contain more than m_n observations and whose depth is less than k_n , there exists a constant C depending on $\|f^*\|_\infty$ such that, for all n large enough,

$$\frac{\log d}{Cm_n} \leq \mathbb{E} \left[\sup_{T \in \mathcal{T}_n(m_n, k_n)} G_0(T) \right] \leq C \frac{k_n \log(nd)}{m_n}, \quad (14)$$

where the lower bound holds when $f^* = 0$ and $m_n \gtrsim \log n$. The population version of the sum of MDIs of irrelevant features is zero, but its empirical counterpart is lower bounded by $(\log d)/Cm_n$. Thus, a part of MDI bias is related to the number of observations in each leaf. Increasing the number of observations per leaf, or equivalently, reducing the depth of the tree can help to diminish this bias. However, it may also decrease the predictive performance of the tree, therefore leading to erroneous MDI.

Two different types of MDI biases have been identified. Biases (related to the splitting criterion) in favor of several types of variables can be reduced by computing MDI on a dataset that was not used during the training phase. Biases due to tree depths (or equivalently, low numbers of observations per leaf) can be reduced by simply limiting the tree depth. These two types of bias act separately: for instance, decreasing tree depth reduces the second bias but has no effect on the first. Note that the standard implementation of RF (i) uses a unique dataset to build trees and compute MDI and (ii) builds fully-grown trees, thus exacerbating MDI biases. Finally, the population value targeted by MDI remains unclear outside of specific generating structures such as generalized additive models (Scornet, 2023).

5.2 Mean Decrease in Accuracy (MDA)

Mean Decrease in Accuracy (MDA, see Breiman, 2001) is built on a random permutation of the values of a given variable in the test set. The difference between the RF predictive performance on this modified test set and on the original test set is called the MDA. MDA is also called permutation importance and can be used for any learner (Fisher et al., 2019), including neural networks (Gevrey et al., 2003). When computing permutation importance within RF, the predictive performance of each tree is measured and the result aggregated, thus making it amenable to the variance estimation methods discussed in Section 4) and allowing tree’s out-of-bag samples to be used in lieu of a test set.

As illustrated by Strobl et al. (2008), MDA is biased in favor of correlated inputs since (i) splits are more likely to be performed along correlated inputs and (ii) permutation importance itself is biased in favor of correlated inputs. To understand this last point, note that if the population MDA is nonzero, then there exists a dependence between $X^{(j)}$ and $(Y, X^{(-j)})$, where $X^{(-j)}$ is the set of all input variables different from $X^{(j)}$. Thus MDA, which is meant to assess the link between $X^{(j)}$ and Y , is biased as soon as $X^{(j)}$ and $X^{(-j)}$ are not independent. Strobl et al. (2008) propose to overcome this limitation by computing conditional permutation importance.

Another limitation of unconditional permutation is due to the nature of the permuted samples. Indeed, [Hooker et al. \(2021\)](#) explain that, because of correlation between inputs, permuting values of a given variable may create unrealistic observations. Thus, MDA evaluates the extrapolation properties of RF rather than the importance of the given variable.

The first theoretical analysis of permutation importance for trees is by [Ishwaran \(2007\)](#). In order to evaluate the importance of a variable $X^{(j)}$, the error of a tree on noisy observations is computed, where the noise comes from randomly assigning the observation to left/right node whenever a split occurs along $X^{(j)}$, and do the same for all subsequent splits in the tree. [Ishwaran \(2007\)](#) establishes the limit of such an importance measure, which resembles the permutation importance. [Gregorutti et al. \(2017\)](#) were the first to provide the population expression of the MDA, in an additive model with Gaussian inputs.

[Bénard et al. \(2022\)](#) establish the convergence of MDA and interpret the limit quantity by the yardstick of sensitivity analysis. Denote the Total Sobol Index of $X^{(j)}$, corresponding to the proportion of explained variance lost when $X^{(j)}$ is removed from the full model, by

$$ST^{(j)} = \frac{\mathbb{E}[\mathbb{V}[f^*(X)|X^{(-j)}]]}{\mathbb{V}[Y]}. \quad (15)$$

Then, considering a random forest composed of M trees with t_n leaves, built with subsamples of size a_n (with $t_n/a_n \rightarrow 0$, $a_n/n \leq 1 - \kappa$ for some $\kappa \in (0, 1)$), [Bénard et al. \(2022\)](#) proves that the MDA of such a forest satisfies

$$\widehat{MDA}_{M,n}(X^{(j)}) \rightarrow \mathbb{V}[Y] \times ST^{(j)} + \mathbb{V}[Y] \times ST_{mg}^{(j)} + MDA_3^{*(j)}, \quad (16)$$

where, letting X_{π_j} the variable X whose j th component is replaced by an independent copy,

$$ST_{mg}^{(j)} = \frac{\mathbb{E}[\mathbb{V}[f^*(X_{\pi_j})|X^{(-j)}]]}{\mathbb{V}[Y]}, \quad (17)$$

and $MDA_3^{*(j)}$ a positive quantity, with a closed-form expression. The limiting quantity in (16) is difficult to interpret, and does not correspond to any known sensitivity index ([Bénard et al., 2022](#)). When performing backward selection, starting from the model with d covariates and eliminating iteratively the variable that decreases the least the predictive performances, the variable to discard is the one with the smallest $ST^{(j)}$. However, it can be proved that the quantity in (16) reduces to $ST^{(j)}$ only in the case of independent features. Such a result highlights the drawback of MDA in presence of correlated inputs.

[Williamson et al. \(2021, 2023\)](#) consider variable importance that can be written as the difference in predictive capacity of a model trained with the full set of input variable compared to that trained without the variable(s) of interest. Computing such measures require retraining the whole model for each variable/group of variables whose importance needs to be measured, which can be computationally intractable. Such measures have been studied for other predictors than random forests by [Lei et al. \(2018\)](#). Other methods to evaluate variable importance have been proposed (see, e.g., Section 5 in [Hooker et al., 2021](#)). [Bénard et al. \(2022\)](#) note that from a forest trained on the full dataset, one can extract a forest trained on a subset of variables, the so-called projected forests. A variable importance can be computed with such forests, which is proved to target the quantity $ST^{(j)}$.

[Ishwaran and Lu \(2019\)](#) (Section 3.2 and 4.4) establish asymptotic confidence intervals of MDA based on subsampling. Theorem 1 in [Williamson et al. \(2023\)](#) establishes the

asymptotic normality of generic importance measures that are computed as the difference in predictive performance between a predictor trained on all variables and the same predictor trained on a strict subset.

One could be tempted to use such results to test whether a variable is important. However, in the presence of a variable of null importance, asymptotic normality does not hold, and a specific test must be designed (see Section 3.4 in [Williamson et al., 2023](#)). [Hooker et al. \(2021\)](#) and [Mentch and Zhou \(2022\)](#) establish that the testing procedure described in [Williamson et al. \(2023\)](#) has a tendency to incorrectly reject the null hypothesis (i.e., to identify irrelevant variables as relevant). Indeed, [Mentch and Zhou \(2022\)](#) prove that adding noise features (independent of the output) can increase the predictive performance of random forests. This is not surprising, as this can be seen as a regularization procedure ([Mentch and Zhou, 2022](#)). However, this questions the interpretation drawn from such tests: the null hypothesis for a variable can be rejected, thus implying increased predictive performances when the variable is added to the model, whereas the variable itself is not related to the output (see the discussion in Section 5.1 in [Mentch and Zhou, 2022](#)).

5.3 Tree SHAP

Both MDI and MDA produce measures of the importance of each input as measured in its contribution to the forest as a whole. Local feature importance asks about how features contribute to making the specific prediction $f_n(x)$ at a given x . A number of model-agnostic feature attribution methods have been proposed based on gradients ([Simonyan et al. \(2014\)](#)), or local sparse linear approximations ([Ribeiro et al. \(2016\)](#)) which have developed a significant literature. Among these, Shapley values [Lundberg and Lee \(2017\)](#) are particularly notable for being able to leverage the structure of trees to significantly improve computation.

Shapley values borrow from game-theoretic considerations for valuing individuals in cooperative games ([Shapley \(1953\)](#)). These partition the total reward from a game among its participants in manner originally designed to satisfy a set of desiderata associated with fair rewards and are based on combining the rewards (or *value function*) that would be obtained for every subset of the participants. In the context of local interpretation, [Lundberg and Lee \(2017\)](#) suggests examining $f_S(x_S)$: the prediction that we would make if we were only given access to the features whose indices are in the set S . Rather than fitting new models this is taken to be the expected value of the prediction over the features not in S :

$$f_S(x_S) = Ef(x_S, X_{-S})$$

where the expectation can be with respect to the marginal distribution of X_{-S} (yielding “interventional” Shapley values) or its conditional distribution $X_{-S}|x_S$ (“conditional” Shapley values), both of which are obtained by Monte-Carlo averaging for a generic f .

Formally, the Shapley value for feature j is then defined by averaging its change to the prediction when it enters the model, with the average taken over permutations of the ordering of features. Letting $\pi = (\pi_1, \dots, \pi_d)$ be a permutation of the feature indices with $S_j^\pi = \{\pi_1, \dots, \pi_{j-1}\}$ the set of indices appearing before j , we define the j th SHAP value as

$$\phi_j = \frac{1}{d!} \sum_{\pi} \left(f_{S_j^\pi \cup \{j\}}(x_{S_j^\pi \cup \{j\}}) - f_{S_j^\pi}(x_{S_j^\pi}) \right). \quad (18)$$

SHAP has become one of the most common feature attribution methods in machine learning, see eg [Holzinger et al. \(2020\)](#), [Molnar \(2020\)](#) although critiques point to both the computa-

tional expense, the rather abstract definition, and a disjunction between the desiderata that were used to define Shapley values and the specific properties need for variable importance (Verdinelli and Wasserman (2024)).

In the context of tree-based models, Shapley values can be computed efficiently based in two properties. First, the conditional expectations involved in calculating $f_S(x_S)$ can be achieved analytically by weighted averages of the leaves; this is particularly simple in the case of interventional Shapley values (see B  nard et al., 2022). Second, the set of permutations to be considered can be substantially reduced: only those π for which some feature used in nodes encountered by x appears after j , and only for the trees for which feature j appears along that path (see Lundberg et al., 2018, Laberge and Pequignot, 2022, for algorithmic details).

Few theoretical studies of Shapley values have been carried out, but we observe that the consistency of a random forest also translates to consistency of the conditional expectations in (18). As noted above, these values are also expressed as averages over values calculated for each tree and for which there is also a central limit theorem.

6 DISCUSSION

The last two decades have seen important advances in random forest theory. Designing and studying random forests variants have led to a better understanding of the role of each ingredient (subsampling, feature randomization, tree depth) on the predictive performance of RF. However, quite notably, the roles of two ingredients remain largely unaddressed: the impact of fully-developed trees and the number of features eligible for splitting. Indeed, they are intrinsically difficult to analyze since many nodes in fully-developed trees has very few observations (while statistical guarantees require many data points) and since analyzing the number of preselected features requires a (data-dependent) manner to select the best split, which is, on its own, a difficulty.

Additionally, a unique RF model that allows for studying possible interactions between these ingredients has yet to be found. Most models are tailored for the analysis of one ingredient only. For example, centered forests are well suited for the analysis of split randomization, but make the analysis of fully-developed trees impossible.

Most analyses assess the impact of RF hyperparameters on predictive performances, but the impact on variable importance remains largely unclear. For example, RF of deep trees have good predictive performances while the resulting variable importances (either MDI or MDA) are clearly biased. Understanding the tradeoff between predictive performance and variable importance (that is, interpretability) is a promising future avenue for research.

Finally, the greatest source of open problems involves understanding adaptively chosen splits. Adaptation is often listed as one of the key advantages of tree-based models, but mathematical results supporting this are limited. While conditions such as regularity and honesty in Wager and Athey (2018) allow for adaptively chosen splits, they are designed to provide a means of ignoring it within the mathematics. A fully developed theory would provide a finer-level description of the adaptation of node geometry to the target model, elucidating the role of adaptive tree kernels both for secondary uses as in Tan et al. (2020) and for combinations of random forest models (Ghosal et al., 2025), as well as controlling the dependence between leaf values and tree structures.

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